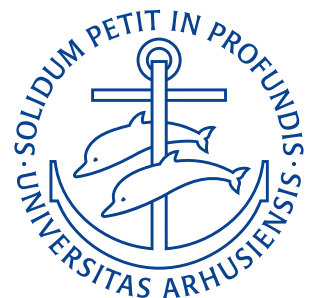


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Lecture Notes

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Lecture 1: Introduction

26. Januar 2026

2 Introduction

This collection of notes are the personal notes of me, Noah Hansen, for the course “Theory of Elasticity” (Da: Elasticitetsteori) taught on the 4th semester of the bachelor on the mechanical engineering programme at Aarhus Universitet.

The notes will take base in the lectures by Assistant Professor Matteo Pezzulla and “Introduction to Linear Elasticity” by Phillip L. Gould and Yuan Feng.

2.1 Coordinate Rotation

Given a position vector $\mathbf{r}$ to a point P , we can resolve it into components with respect to two different Cartesian systems, x_i and x'_i , having a common origin.

The point P has coordinates $P(x_1, x_2, x_3) = P(x_i)$ in the unprimed system and $P(x'_1, x'_2, x'_3) = P(x'_i)$ in the primed system. The linear transformation between the coordinates is then:

$$\begin{aligned}x'_1 &= \alpha_{11}x_1 + \alpha_{12}x_2 + \alpha_{13}x_3 \\x'_2 &= \alpha_{21}x_1 + \alpha_{22}x_2 + \alpha_{23}x_3 \\x'_3 &= \alpha_{31}x_1 + \alpha_{32}x_2 + \alpha_{33}x_3\end{aligned}$$

or

$$x'_i = \alpha_{ij}x_j.$$

Each of the nine quantities α_{ij} is the cosine of the angle between the i th primed and the j th unprimed axis, i.e.

$$\alpha_{ij} = \cos(x'_i, x_j) = \frac{\partial x_j}{\partial x'_i} = \mathbf{e}'_i \cdot \mathbf{e}_j = \cos(\mathbf{e}'_i, \mathbf{e}_j).$$

These α_{ij} 's are known as direction cosines.

2.2 Cartesian Tensors

A tensor of order n is a set of 3^n quantities which transform from one coordinate system x_i to another x'_i , by the following law:

Table 1: Tranformation laws for Tensors

n	Order	Transformation law
0	Zero (scalar)	$A'_i = A_i$
1	One (vector)	$A'_i = \alpha_{ij}A_j$
2	Two (dyadic)	$A'_{ij} = \alpha_{ik}\alpha_{jl}A_{kl}$
3	Three	$A'_{ijk} = \alpha_{il}\alpha_{jm}\alpha_{kn}A_{lmn}$
4	Four	$A'_{ijkl} = \alpha_{im}\alpha_{jn}\alpha_{kp}\alpha_{lq}A_{mnpq}$

Order zero and order one tensors (scalars and vectors are already familiar quantities, wheres higher-order tensors are not. These tensors are, however, useful to describe physical quantities with a corresponding number of associated directions. Second-order tensors (dyadics) are especially prevalent for elasticity and the corresponding transformation may be carried out in matrix format as:

$$\mathbf{A}' = \mathbf{R}\mathbf{A}\mathbf{R}^T.$$

in which

$$\mathbf{R} = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \alpha_{13} \\ \alpha_{21} & \alpha_{22} & \alpha_{23} \\ \alpha_{31} & \alpha_{32} & \alpha_{33} \end{bmatrix} \quad \text{and} \quad \mathbf{A}' = \begin{bmatrix} A'_{11} & A'_{12} & A'_{13} \\ A'_{21} & A'_{22} & A'_{23} \\ A'_{31} & A'_{32} & A'_{33} \end{bmatrix}.$$

2.2.1 Algebra of Cartesian Tensors

Arithmetic and algebra for tensors is similar to matrix operations when it comes to addition, subtraction, equality, and scalar multiplication. Multiplication of two tensors of order n and m produces a new tensor of order $n + m$, e.g.

$$A_i B_{jk} = C_{ijk}.$$

2.3 Operational Tensors

Additional tensor operations are made possible by the Kronecker delta δ_{ij} , defined as,

$$\delta_{ij} \equiv \begin{cases} 1 & i = j \\ 0 & i \neq j. \end{cases}$$

and the permutation tensor ϵ_{ijk} defined such that:

$$\epsilon_{ijk} = \frac{1}{2}(i-j)(j-k)(k-i)$$

$$= \begin{cases} 0 & \text{If any two of } i, j, k \text{ are equal} \\ 1 & \text{For an even permutation (forward on the number line } 1, 2, 3, 1, 2, 3, \dots) \\ -1 & \text{For an odd permutation (backward on the number line } 3, 2, 1, 3, 2, 1, \dots). \end{cases}$$

The Kronecker delta δ_{ij} changes the subscripts in a tensor by multiplication:

$$\delta_{ij} A_i = \delta_{1j} A_1 + \delta_{2j} A_2 + \delta_{3j} A_3 = A_j$$

$$\delta_{ij} C_{ijk} = \delta_{11} C_{11k} + \delta_{22} C_{22k} + \delta_{33} C_{33k} = C_{iik} = A_K.$$

It is also useful in vector algebra and for coordinate transformations. E.g. starting with the dot product of two unit vectors:

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij} \tag{1}$$

and seeking the component of vector $\mathbf{A}$ in the j -direction A_j we do:

$$\begin{aligned} \mathbf{e}_j \cdot \mathbf{A} &= (\mathbf{e}_j \cdot \mathbf{e}_i) A_i \\ &= \delta_{ji} A_i \\ &= A_j. \end{aligned}$$

The permutation tensor ϵ_{ijk} is useful for vector cross-product operations:

$$\epsilon_{ijk} A_j B_k \mathbf{e}_i = \mathbf{A} \times \mathbf{B}.$$

Lecture 2: Traction, Stress, and Equilibrium

2. Februar 2026

4 Traction, Stress, and Equilibrium

4.1 State of Stress

For a deformable body subject to external loading we focus on an element with area ΔA_n oriented according to the unit normal $\mathbf{n}$, we call the resultant force $\Delta \mathbf{F}_n$ and the resultant moment $\Delta \mathbf{M}_n$. Both are vector quantities, and are *not*, in general, parallel to $\mathbf{n}$. We represent the intensity of these as

$$\lim_{\Delta A_n \rightarrow 0} \frac{\Delta \mathbf{F}_n}{\Delta A_n} = \frac{d\mathbf{F}_n}{dA_n} = \mathbf{T}_n$$

$$\lim_{\Delta A_n \rightarrow 0} \frac{\Delta \mathbf{M}_n}{\Delta A_n} = \frac{d\mathbf{M}_n}{dA_n} = \mathbf{C}_n$$

where $\mathbf{T}_n$ is known as the stress vector or the *traction*, and $\mathbf{C}_n$ is called the *couple-stress vector*.

4.2 Components of Stress

We consider an infinitesimal rectangular prism erected at the point from [Section 4.1](#) and erect a set of Cartesian coordinates x_i parallel to the sides as shown on [Figure 4.1](#). Each of these coordinate axis has a corresponding unit vector $\mathbf{e}_i$.

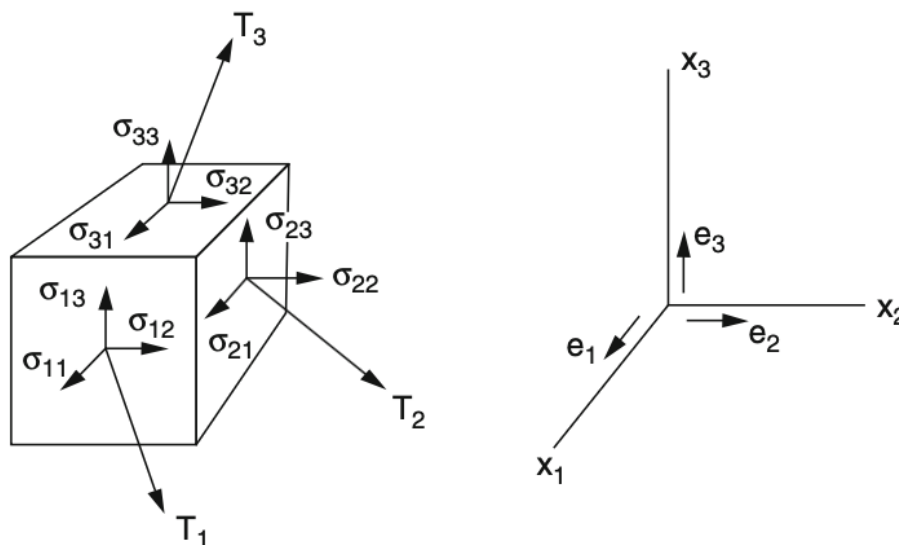


Figure 4.1: Components of stress.

Each traction may be written in terms of its Cartesian components in the form

$$\mathbf{T}_i = \sigma_{ij} \mathbf{e}_j \quad (2)$$

which is shown for $\mathbf{T}_1$ below

$$\mathbf{T}_1 = \sigma_{11} \mathbf{e}_1 + \sigma_{12} \mathbf{e}_2 + \sigma_{13} \mathbf{e}_3.$$

In [Equation \(2\)](#), the coefficients σ_{ij} are known as stresses, and the entire array of stresses forms the so-called Cauchy stress tensor. The subscript and sign convention for stress components σ_{ij} are:

1. The first subscript i refers to the *normal* $\mathbf{e}_i$, which denotes the face on which $\mathbf{T}_i$ acts.
2. The second subscript j corresponds to the *direction* $\mathbf{e}_j$ in which the stress acts.

3. The normal components σ_{ii} are positive if they produce tension and negative if they produce compression. The shearing components $\sigma_{ij} (i \neq j)$ are positive if directed in the positive x_j direction while acting on the face with the unit normal e_i or if directed in the negative x_j direction while acting on the face with normal $-e_i$

4.3 Stress at a Point

It can be shown that to completely describe the state of stress at a point, it is sufficient to specify the tractions T_i on each of the three planes e_i – i.e. this is equivalent to specifying the nine stress components σ_{ij} .

$$T_i = \sigma_{ji} n_j. \quad (3)$$

And if the components T_i are known the magnitude of T_n is evaluated as

$$T_n = |T_n| = (T_i T_i)^{\frac{1}{2}}.$$

4.3.1 Stress on a Normal Plane

It is sometimes useful to resolve T_n into components that are normal and tangential to the differential surface element dA_n as depicted on Figure 4.2. Here, the normal component σ_{nn} is calculated by

$$\begin{aligned} \sigma_{ns} &= |N| = T_n \cdot n \\ &= T_i e_i \cdot n = T_i n_i. \end{aligned}$$

or using Equation (3) as

$$\sigma_{nn} = \sigma_{ji} n_j n_i. \quad (4)$$

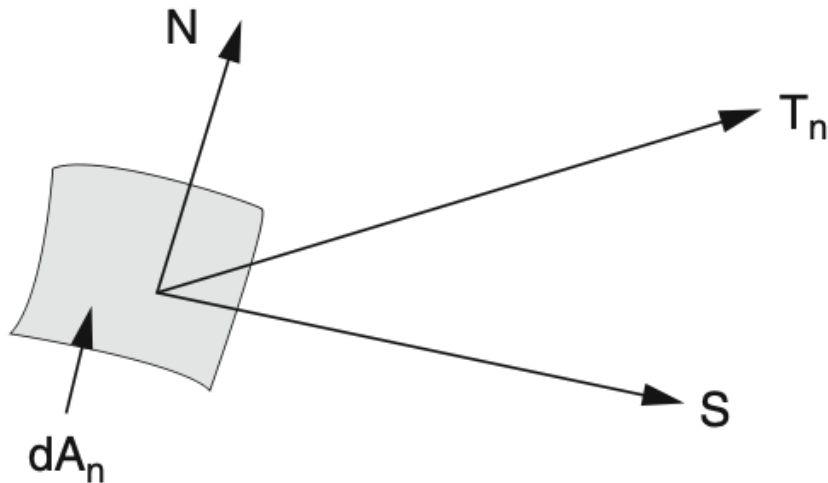


Figure 4.2: Differential surface element.

The tangential component σ_{ns} is

$$\begin{aligned} \sigma_{ns} &= |S| = T_n \cdot s \\ &= T_i e_i \cdot s \\ &= T_i s_i \\ &= \sigma_{ji} n_j s_i. \end{aligned}$$

where

$$s_i = \mathbf{e}_i \mathbf{s}.$$

Often it is easy to calculate σ_{ns} using the Pythagorean theorem as

$$\sigma_{ns} = \sqrt{T_i T_i - \sigma_{nn}^2}.$$

One is also able to resolve N and S into Cartesian components using (k) to designate the axes as

$$\begin{aligned}\sigma_{nn(k)} &= \mathbf{N} \cdot \mathbf{e}_k = \sigma_{nn} \mathbf{n} \cdot \mathbf{e}_k \\ &= \sigma_{nn} n_k \\ &= \sigma_{ji} n_j n_i n_k, \quad k = 1, 2, 3.\end{aligned}$$

For σ_{ns} subtraction gives

$$\sigma_{ns(k)} = T_k - \sigma_{nn(k)} \quad k = 1, 2, 3$$

which may also be written as

$$\sigma_{ns(k)} = \sigma_{jk} n_j - \sigma_{ji} n_j n_i n_k = (\delta_{ki} - n_i n_k) \sigma_{ji} n_j.$$

4.3.2 Dyadic Representation of Stress

One is able to view the stress tensor as a vector-like quantity with a magnitude and associated direction(s) specified by unit vectors. The *dyadic* is such a representation. We write the stress tensor or stress dyadic as

$$\begin{aligned}\boldsymbol{\sigma} &= \sigma_{ij} \mathbf{e}_i \mathbf{e}_j \\ &= \sigma_{11} \mathbf{e}_1 \mathbf{e}_1 + \sigma_{12} \mathbf{e}_1 \mathbf{e}_2 + \sigma_{13} \mathbf{e}_1 \mathbf{e}_3 + \sigma_{21} \mathbf{e}_2 \mathbf{e}_1 + \sigma_{22} \mathbf{e}_2 \mathbf{e}_2 + \sigma_{23} \mathbf{e}_2 \mathbf{e}_3 + \sigma_{31} \mathbf{e}_3 \mathbf{e}_1 + \sigma_{32} \mathbf{e}_3 \mathbf{e}_2 + \sigma_{33} \mathbf{e}_3 \mathbf{e}_3\end{aligned}$$

wherein the double vectors are called *dyads*. The corresponding tractions are evaluated by an operation analogous to the dot product:

$$\mathbf{T}_i = \boldsymbol{\sigma} \cdot \mathbf{e}_i = \sigma_{ij} \mathbf{e}_j.$$

Similarly, the normal and tangential components of the traction $\mathbf{T}_n$ on a plane defined by normal $\mathbf{n}$ are

$$\begin{aligned}\sigma_{nn} &= \boldsymbol{\sigma} \cdot \mathbf{n} \cdot \mathbf{n} \\ &= \mathbf{T}_n \cdot \mathbf{n} \\ &= \sigma_{ij} n_i n_j\end{aligned}$$

and

$$\begin{aligned}\sigma_{ns} &= \boldsymbol{\sigma} \cdot \mathbf{n} \cdot \mathbf{s} \\ &= \mathbf{T}_n \cdot \mathbf{s} \\ &= \sigma_{ij} n_i s_j.\end{aligned}$$

Lecture 3: The Cauchy theorem and principal stresses

9. Februar 2026

Lecture 4: Equilibrium Equations & Principle of Angular Momentum

16. Februar 2026

6.1 Equilibrium**6.1.1 Foundations**

The principle of linear momentum is

$$\int_V \mathbf{f} \, dV + \int_A \mathbf{T} \, dA = \int_V \rho \ddot{\mathbf{u}} \, dV \quad (5)$$

where ρ is the mass density and $\mathbf{u}$ is the displacement.

The principle of angular momentum is

$$\int_V (\mathbf{r} \times \mathbf{f}) \, dV + \int_A (\mathbf{r} \times \mathbf{T}) \, dA = \int_V (\mathbf{r} \times \rho \ddot{\mathbf{u}}) \, dV \quad (6)$$

where $\mathbf{r}$ is the position vector.

We will also utilize the divergence theorem:

$$\int_V \nabla \cdot \mathbf{G} \, dV = \int_A \mathbf{n} \cdot \mathbf{G} \, dA. \quad (7)$$

These can each be written in component form using:

$$\begin{aligned} \mathbf{f} &= f_i \mathbf{e}_i \\ \mathbf{T} &= \sigma_{ji} n_j \mathbf{e}_i \\ \mathbf{r} &= x_j \mathbf{e}_j \\ \mathbf{r} \times \mathbf{f} &= \epsilon_{ijk} x_j f_k \mathbf{e}_i \\ \mathbf{r} \times \mathbf{T} &= \epsilon_{ijk} x_j \sigma_{lk} n_l \mathbf{e}_i. \end{aligned}$$

For static problems, the rhs of [Equations \(5\) to \(7\)](#) are zero. Substituting this and using the above component-form we get

$$\int_V f_i \, dV + \int_A \sigma_{ji} n_j \, dA = 0 \quad (8)$$

$$\int_V \epsilon_{ijk} x_j f_k \, dV + \int_A \epsilon_{ijk} x_j \sigma_{lk} n_l \, dA = 0 \quad (9)$$

$$\int_V G_{i,i} \, dV = \int_A n_i G_i \, dA. \quad (10)$$

6.1.2 Linear Momentum

If we apply [Equation \(10\)](#) with $G_i = \sigma_{ji}$ to [Equation \(8\)](#) we get:

$$\int_V f_i + \sigma_{ji,j} \, dV = 0.$$

For all elements of V to be in equilibrium, the above equation is satisfied if the integrand vanishes, i.e.

$$\sigma_{ji,j} + f_i = 0. \quad (11)$$

This represents three equilibrium equations in terms of nine unknown components of stress σ_{ij} .

6.1.3 Angular Momentum

We apply Equation (10) with $G_i = \epsilon_{ijk} x_j \sigma_{lk}$ and $n_i = n_l$ to Equation (9) and get:

$$\int_V \epsilon_{ijk} x_j f_k + (\epsilon_{ijk} x_j \sigma_{lk})_{,l} dV = 0. \quad (12)$$

The second term is expanded as

$$(\epsilon_{ijk} x_j \sigma_{lk})_{,l} = \epsilon_{ijk} \sigma_{lk} x_{j,l} + \epsilon_{ijk} x_j \sigma_{lk,l}.$$

Also we use that x_j only changes in the j 'th direction: i.e.

$$x_{j,l} = \delta_{jl}.$$

Hence, Equation (12) becomes:

$$\int_V \epsilon_{ijk} (x_j f_k + \delta_{jl} \sigma_{lk} + x_j \sigma_{lk,l}) dV = 0$$

the first and third term combines to

$$\epsilon_{ijk} x_j (f_k + \sigma_{lk,l}) = 0$$

from Equation (11). The remaining term becomes

$$\epsilon_{ijk} \delta_{jl} \sigma_{lk} = \epsilon_{ijk} \sigma_{jk}.$$

So the equation reduces to

$$\int_V \epsilon_{ijk} \sigma_{jk} dV = 0$$

which holds for

$$\epsilon_{ijk} \sigma_{jk} = 0.$$

In general, this is satisfied for

$$\sigma_{ij} = \sigma_{ji}.$$

This allows us to rewrite the component definition of traction as

$$T_i = \sigma_{ij} n_j$$

and Equation (11) as

$$\sigma_{ij,j} + f_i = 0.$$

Which corresponds to the statically indeterminate system:

$$\sigma_{11,1} + \sigma_{12,2} + \sigma_{13,3} + f_1 = 0$$

$$\sigma_{21,1} + \sigma_{22,2} + \sigma_{23,3} + f_2 = 0$$

$$\sigma_{31,1} + \sigma_{32,2} + \sigma_{33,3} + f_3 = 0.$$

6.2 Principal Stress

6.2.1 Definition and Derivation

We consider the normal section through a body shown on Figure 6.1a. The traction T_n acts on the plane defined by $\mathbf{n}$ that appears edgewise.

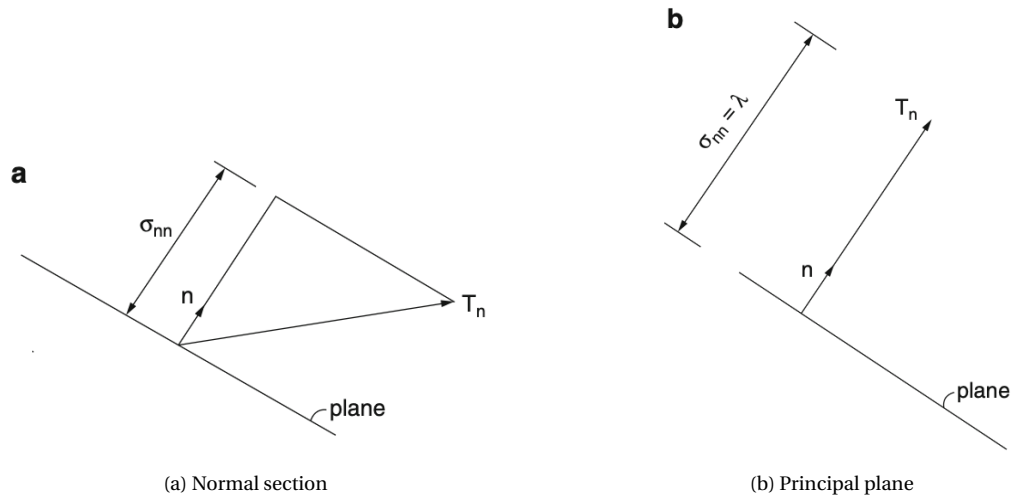


Figure 6.1: Normal section and principal plane

Using the dyadic form,

$$\sigma_{nn} = \mathbf{T}_n \cdot \mathbf{n} = \boldsymbol{\sigma} \cdot \mathbf{n} \cdot \mathbf{n}$$

we seek the orientation of a plane, given by $\mathbf{n}$, such that σ_{nn} is an extremum. Such a plane is called a *principal* plane.

Hence,

$$\begin{aligned} \frac{d\sigma_{nn}}{dn} &= \mathbf{n} \cdot \frac{d(\boldsymbol{\sigma} \cdot \mathbf{n})}{dn} + (\boldsymbol{\sigma} \cdot \mathbf{n}) \cdot \frac{d\mathbf{n}}{dn} \\ &= 2(\boldsymbol{\sigma} \cdot \mathbf{n}) \cdot \frac{d\mathbf{n}}{dn} \\ &= 2\mathbf{T}_n \cdot \frac{d\mathbf{n}}{dn} \\ &= 0. \end{aligned}$$

For $\mathbf{T}_n \cdot \frac{d\mathbf{n}}{dn}$ to equal 0, $\mathbf{T}_n$ must be normal to $\frac{d\mathbf{n}}{dn}$. Furthermore, $\frac{d\mathbf{n}}{dn}$ is normal to $\mathbf{n}$, so $\mathbf{T}_n$ must be parallel to $\mathbf{n}$. This condition corresponding to the extremum is shown in [Figure 6.1b](#). Here, the tangential component $\sigma_{ns} = 0$ and the normal component for this case is designated as

$$\sigma_{nn} = |\mathbf{T}_n| = \lambda$$

which may also be represented as

$$\mathbf{T}_n = \lambda \mathbf{n}$$

or in component form as

$$T_i = \lambda n_i.$$

Also, we know that

$$T_i = \sigma_{ji} n_j.$$

Equating these gives

$$\lambda n_i = \lambda n_j \delta_{ij} = \sigma_{ji} n_j$$

or

$$(\sigma_{ji} - \lambda \delta_{ij}) n_j = 0. \quad (13)$$

This is a set of three equations in four unknowns, n_1, n_2, n_3 , and λ . The components n_j determine the orientation of the principal plane and λ is the principal stress. This is a linear eigenvalue problem, and with Cramer's rule, it can be shown that a nontrivial solution may be found if the determinant of the coefficients of n_j vanishes, i.e.

$$|\sigma_{ji} - \lambda \delta_{ij}| = 0$$

or

$$\begin{vmatrix} \sigma_{11} - \lambda & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} - \lambda & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} - \lambda \end{vmatrix} = 0.$$

This expands into a cubic equation for λ indicating that three principal stresses, $\lambda_1 = \sigma^{(1)}, \lambda_2 = \sigma^{(2)}, \lambda_3 = \sigma^{(3)}$, exist. For each principal stress, a corresponding distinct principal orientation may be evaluated by substituting, a known value of $\sigma^{(k)}$ into Equation (13). As this still represents three equations in four unknowns, the additional supplementary relationship:

$$n_j^{(k)} n_j^{(k)} = 1$$

expressing the “length” of the unit normal for $k = 1, 2, 3$ must also be introduced.

6.3 Properties and Special States of Stress

6.3.1 Plane Stress

Consider a plane passing through P with no stresses, i.e. $T = 0$. The projection theorem states that the traction on any other plane passing through P must be perpendicular to the normal to the stress-free plane; hence, it is parallel to that plane. Consequently, if the traction on a plane is parallel to another plane, then the latter plane will be stress free. This is known as a state of *plane stress* and is of practical use since this allows us to reduce the dimensionality of elasticity problems.

If one of the principal stresses is zero, the state of stress is plane.

6.3.2 Linear Stress

For two nonparallel planes on which $T = 0$, the traction on any other plane must be parallel to both of these planes and also to their line of intersection. This is a state of *linear stress* and is essentially a one-dimensional problem, where two principal stresses vanish.

6.3.3 Pure Stress

If, for any coordinate system

$$\sigma_{11} = \sigma_{22} = \sigma_{33} = 0$$

the state of stress is called *pure shear* since only shear stresses, $\sigma_{ij}, i \neq j$ exist. For pure shear, $\sigma_{ii} = 0$ for any coordinate system.

6.3.4 Hydrostatic Stress

If the principal stresses $\sigma^{(1)}$ are all equal, then the stress state is hydrostatic, and the shearing stresses are zero.

Lecture 5: The Strain Tensor

23. Februar 2026

8 Deformations

8.1 Strain

Consider the undeformed and deformed positions of an elastic body shown on Figure 8.1. It is convenient to designate two sets of Cartesian coordinates x_i and X_i with $i = 1, 2, 3$, called *initial* coordinates and *final* coordinates, respectively. Now, we designate a reference point $P(\bar{x}_i)$ on the undeformed body and $p(\bar{X}_i)$ on the deformed body, where $\bar{x}_i$ and $\bar{X}_i$ indicate specific values of the coordinates x_i and X_i . Also we locate the neighbouring points $Q(\bar{x}_i + d\bar{x}_i)$ and $q(\bar{X}_i + d\bar{X}_i)$, separated from P and p by ds and dS respectively. The latter are given in component form by

$$(ds)^2 = d\bar{x}_1^2 + d\bar{x}_2^2 + d\bar{x}_3^2 = d\bar{x}_i d\bar{x}_i$$

$$(dS)^2 = d\bar{X}_1^2 + d\bar{X}_2^2 + d\bar{X}_3^2 = d\bar{X}_i d\bar{X}_i.$$

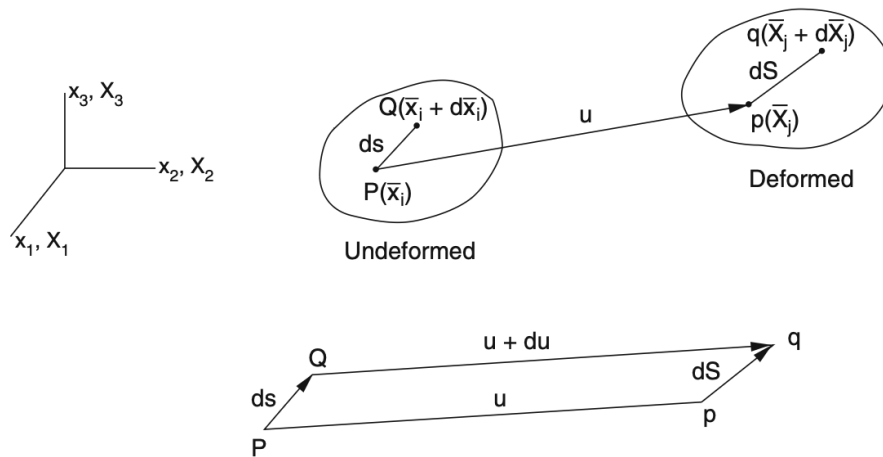


Figure 8.1: Displacement vector

In Figure 8.1, u represents the displacement of P to p and $u + du$ is the displacement of Q to q . From the vector diagram we note

$$ds + (u + du) = u + dS \implies du = dS - ds.$$

We consider the relationship in component form:

$$(dS)^2 - (ds)^2 = d\bar{X}_i d\bar{X}_i - d\bar{x}_i d\bar{x}_i.$$

Now, we only consider the initial coordinates x_i in this next step, so that

$$\begin{aligned} \bar{X}_i &= \bar{X}_i(x_1, x_2, x_3) \\ d\bar{X}_i &= \frac{\partial X_i}{\partial x_1} d\bar{x}_1 + \frac{\partial X_i}{\partial x_2} d\bar{x}_2 + \frac{\partial X_i}{\partial x_3} d\bar{x}_3 \\ &= X_{i,j} d\bar{x}_j \\ \mathbf{u} &= u_1 \mathbf{e}_{x_1} + u_2 \mathbf{e}_{x_2} + u_3 \mathbf{e}_{x_3} = u_i \mathbf{e}_{x_i} \\ u_i &= X_i - x_i. \end{aligned}$$

Applying this to the component form relationship:

$$\begin{aligned}
 (dS)^2 - (ds)^2 &= X_{i,j} d\bar{x}_j X_{i,k} d\bar{x}_k - d\bar{x}_i d\bar{x}_i \\
 &= (X_{i,j} X_{i,k} - \delta_{ij} \delta_{ik}) d\bar{x}_j d\bar{x}_k \\
 &= ((x_i + u_i)_{,j} (x_i + u_i)_{,k} - \delta_{jk}) d\bar{x}_j d\bar{x}_k \\
 &= ((\delta_{ij} + u_{i,j}) (\delta_{ik} + u_{i,k}) - \delta_{ik}) d\bar{x}_j d\bar{x}_k \\
 &= (\delta_{jk} + u_{j,k} + u_{k,j} + u_{i,j} u_{i,k} - \delta_{jk}) d\bar{x}_j d\bar{x}_k \\
 &= (u_{j,k} + u_{k,j} + u_{i,j} u_{i,k}) d\bar{x}_j d\bar{x}_k \\
 &= 2\epsilon_{ij}^L d\bar{x}_i d\bar{x}_j
 \end{aligned}$$

in which

$$\epsilon_{ij}^L = \frac{1}{2} (u_{i,j} + u_{j,i} + u_{k,i} u_{k,j})$$

are components of the Lagrangian strain tensor ϵ^L .

Doing the same for the final coordinates X_i as the basis we obtain

$$\begin{aligned}
 (dS)^2 - (ds)^2 &= (u_{i,j} + u_{j,i} - u_{k,i} u_{k,j}) d\bar{X}_i d\bar{X}_j \\
 &= 2\epsilon_{ij}^E d\bar{X}_i d\bar{X}_j
 \end{aligned}$$

where

$$\epsilon_{ij}^E = \frac{1}{2} (u_{i,j} + u_{j,i} - u_{k,i} u_{k,j})$$

are components of the *Eulerian* strain tensor ϵ^E .

If the displacement gradients $u_{k,i}$ are small, then their products are so too and may be dropped. For such cases:

$$\epsilon_{ij}^L = \epsilon_{ij}^E = \epsilon_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i}).$$

The components ϵ_{ij} refer to the small strain tensor ϵ .

8.2 Physical Interpretation of Strain Tensor

Consider the components $\epsilon_{(ii)}$ called *extensional* strains. E.g. a fiber parallel to x_2 with initial length ds and final length dS , giving $dx_1 = dx_2 = 0$ and $dx_2 = ds$. Hence,

$$(dS)^2 - (ds)^2 = 2\epsilon_{22} ds^2$$

for the case of infinitesimal strain.

This gives:

$$\begin{aligned}
 \frac{dS - ds}{ds} &= \frac{2\epsilon_{22} ds^2}{ds (dS + ds)} \\
 &= \frac{2\epsilon_{22} ds^2}{2ds^2} \\
 &= \epsilon_{22} \\
 \Rightarrow \epsilon_{22} &= \frac{dS - ds}{ds}
 \end{aligned}$$

which is the relative elongation of the fiber.

The components ϵ_{ij} are called *shearing strains* and may be interpreted by considering intersecting fibers initially parallel to two of the coordinate axes. E.g. the x_1 - and x_3 -axes as shown in Figure 8.2. These fibers have initial lengths $ds^{(1)}$ and $ds^{(3)}$, final length $dS^{(1)}$ and $dS^{(3)}$ and have rotated through angles $\beta^{(1)}$ and $\beta^{(3)}$, respectively. The total change in the initially right angle is

$$\beta^{(1)} + \beta^{(3)} = \beta.$$

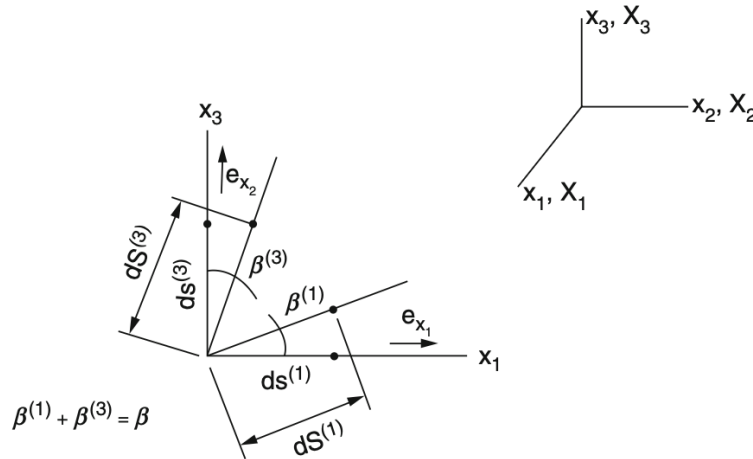


Figure 8.2: Shearing strains

Vectors coinciding with the fibers can be written in component form as:

$$dS^{(1)} = dS_i^{(1)} e_{x_i}$$

and

$$dS^{(3)} = dS_i^{(3)} e_{x_i}.$$

Using these, we can show that

$$\beta = (u_{1,3} + u_{3,1}) = 2\epsilon_{13}.$$

I.e. the strain component ϵ_{13} is one half the increase in the angle between two fibers initially parallel to the x_1 - and x_3 -axes.

Since the strain tensor is formed from the displacement gradients, it is interesting to separate $u_{i,j}$ into two parts:

$$u_{i,j} = \epsilon_{ij} + \omega_{ij}$$

where

$$\omega_{ij} = \frac{1}{2} (u_{i,j} - u_{j,i}).$$

For $i = j$, $\omega_{ij} = 0$, while for $i \neq j$, $\omega_{ij} = -\omega_{ji}$; thus, the ω_{ij} are elements of a skew-symmetric tensor ω . Using the example from before,

$$\omega_{13} = \frac{1}{2} (u_{1,3} - u_{3,1}).$$

Also

$$\beta = \beta^{(1)} + \beta^{(3)} = (u_{1,3} + u_{3,1}) = 2\epsilon_{13}.$$

By comparison we get

$$\omega_{13} = \frac{\beta^{(1)} - \beta^{(3)}}{2}$$

which can be interpreted as the average rotation of all fibers around the x_2 axis. For a rigid body rotation, $\epsilon_{13} = 0$, $\beta^{(1)} = -\beta^{(3)}$, and $\omega_{13} = \beta^{(1)}$. Hence, ω is called the rotation tensor.

Lecture 6: Physical interpretation of the strain tensor

2. Marts 2026

9.1 Principal Strain

As the strains have the same second-order tensor character as the stresses, it is expected that we are able to identify a system of principal strains that are the eigenvalues, determined from the characteristic equation

$$|\epsilon_{ji} - \lambda \delta_{ji}| = 0$$

which is analogous to the result for stresses. The three roots λ_1 , λ_2 , and λ_3 are the principal strains $\epsilon^{(k)}$ with $k = 1, 2$, and 3 .

The corresponding eigenvectors designate the directions associated with each of the principal strains and are computed from

$$(\epsilon_{ji} - \lambda \delta_{ji}) n_j = 0.$$

The directions of principal strain are denoted by $n_j^{(k)}$, with $k = 1, 2$, and 3 . These directions are mutually perpendicular and, for isotropic materials, coincide with the directions of the principal stresses. Which makes logical sense, as the largest stress is also expected to cause the largest strain.

9.2 Volume and Shape Change

Sometimes it is convenient to separate the components of strain into those that cause changes in volume and those that cause change in the shape of a differential element.

Considering a volume element oriented with the principal directions and with shearing stresses equal to zero. The original length of each side is $dx^{(k)}$ and the final length as $dX^{(k)}$ where

$$dX^{(k)} = dx^{(k)} (1 + \epsilon^{(k)}).$$

The original volume is found as

$$dv = \prod_{k=1}^3 dx^{(k)}$$

and the final volume as

$$\begin{aligned} dV &= \prod_{k=1}^3 dX^{(k)} \\ &= \prod_{k=1}^3 (1 + \epsilon^{(k)}) dx^{(k)} \\ &= dv \prod_{k=1}^3 (1 + \epsilon^{(k)}). \end{aligned}$$

We define the *dilation* as the relative change in volume

$$\begin{aligned}
 \Delta &= \frac{dV - dv}{dv} \\
 &= \frac{dv \left(\left(\prod_{k=1}^3 (1 + \epsilon^{(k)}) \right) - 1 \right)}{dv} \\
 &= \left((1 + \epsilon^{(1)}) (1 + \epsilon^{(2)}) (1 + \epsilon^{(3)}) \right) - 1 \\
 &\approx \epsilon^{(1)} + \epsilon^{(2)} + \epsilon^{(3)} \\
 &= \prod_{k=1}^3 \epsilon^{(k)}
 \end{aligned}$$

which is the sum of the diagonal terms of the principal strain tensor.

The dilation Δ is also equal to the sum of the diagonal strains ϵ_{ii} , and the divergence of the displacement vector $\nabla \cdot \mathbf{u} = u_{i,j}$.

Changes in shape of the volume element are called *detrusions* and are denoted by shearing strains $\epsilon_{ij}, i \neq j$.

The separation of the strains into dilational and detrusional components can be formalized by resolving the strain tensor into two terms:

$$\boldsymbol{\epsilon} = \boldsymbol{\epsilon}_M + \boldsymbol{\epsilon}_D$$

where

$$\begin{aligned}
 \boldsymbol{\epsilon}_M &= \begin{bmatrix} \epsilon_M & 0 & 0 \\ 0 & \epsilon_M & 0 \\ 0 & 0 & \epsilon_M \end{bmatrix} = \text{mean normal strain} \\
 \boldsymbol{\epsilon}_D &= \begin{bmatrix} \epsilon_{11} - \epsilon_M & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{12} & \epsilon_{22} - \epsilon_M & \epsilon_{23} \\ \epsilon_{13} & \epsilon_{23} & \epsilon_{33} - \epsilon_M \end{bmatrix} = \text{strain deviator}
 \end{aligned}$$

and where

$$\begin{aligned}
 \epsilon_M &= \frac{1}{3} (\epsilon_{11} + \epsilon_{22} + \epsilon_{33}) \\
 &= \frac{1}{3} (\epsilon^{(1)} + \epsilon^{(2)} + \epsilon^{(3)}) \\
 &= \frac{1}{3} R_1.
 \end{aligned}$$

The mean normal strain corresponds to a state of equal elongation in all directions for an element at a given point. Only subject to this, the element would maintain original shape but change in volume with dilation ϵ_{ii} . Hence, ϵ_M is the *volumetric* component of strain.

The strain deviator, or *deviatoric* component, ϵ_D , has a dilation of

$$\begin{aligned}
 \Delta &= \prod_{i=1}^3 \epsilon_D^{(i)} \\
 &= \epsilon_{Dii} \\
 &= (\epsilon_{11} - \epsilon_M) + (\epsilon_{22} - \epsilon_M) + (\epsilon_{33} - \epsilon_M) \\
 &= \epsilon_{11} + \epsilon_{22} + \epsilon_{33} - 3 \left(\frac{1}{3} (\epsilon_{11} + \epsilon_{22} + \epsilon_{33}) \right) \\
 &= 0.
 \end{aligned}$$

This corresponds to a change in shape with no change in volume of an element.

Corresponding to this, we can also define the principal stresses associated with changes in the volume and shape as

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_M + \boldsymbol{\sigma}_D$$

defined similarly as the stress, where

$$\boldsymbol{\sigma}_M = \begin{bmatrix} \sigma_M & 0 & 0 \\ 0 & \sigma_M & 0 \\ 0 & 0 & \sigma_M \end{bmatrix} = \text{mean normal stress}$$

$$\boldsymbol{\sigma}_D = \begin{bmatrix} \sigma_{11} - \sigma_M & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} - \sigma_M & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} - \sigma_M \end{bmatrix} = \text{stress deviator}$$

and where

$$\begin{aligned} \sigma_M &= \frac{1}{3} (\sigma_{11} + \sigma_{22} + \sigma_{33}) \\ &= \frac{1}{3} (\sigma^{(1)} + \sigma^{(2)} + \sigma^{(3)}). \end{aligned}$$

In indicial form, the volumetric and deviatoric components of stress are, respectively,

$$\begin{aligned} \sigma_{Mij} &= \frac{1}{3} \sigma_{kk} \delta_{ij} \\ \sigma_{Dij} &= \sigma_{ij} - \sigma_{Mij} \\ &= \sigma_{ij} - \frac{1}{3} \sigma_{kk} \delta_{ij}. \end{aligned}$$

For principal axes, the stress deviator becomes

$$\begin{aligned} \boldsymbol{\sigma}_D &= \begin{bmatrix} \sigma^{(1)} - \sigma_M & 0 & 0 \\ 0 & \sigma^{(2)} - \sigma_M & 0 \\ 0 & 0 & \sigma^{(3)} - \sigma_M \end{bmatrix} \\ &= \begin{bmatrix} \sigma_D^{(1)} & 0 & 0 \\ 0 & \sigma_D^{(2)} & 0 \\ 0 & 0 & \sigma_D^{(3)} \end{bmatrix}. \end{aligned}$$

Worth noting here is that the sum of the diagonals of $\boldsymbol{\sigma}_D$ is zero as

$$\sigma_{Dii} = \sigma^{(1)} + \sigma^{(2)} + \sigma^{(3)} - 3 \cdot \frac{1}{3} (\sigma^{(1)} + \sigma^{(2)} + \sigma^{(3)}) = 0.$$

Lecture 7: Volume and shape change

9. Marts 2026

10.1 Compatibility

Using $\epsilon_{ij} = 1/2 (u_{i,j} + u_{j,i})$, the components of strain may be easily computed once displacements are known provided they are once differentiable. However, the inverse problem of calculating displacements from strains is not as straight forward as there are six independent equations with only three unknowns. Conditions of compatibility imposed on the components of strain ϵ_{ij} , are both necessary and sufficient to ensure a continuous single-valued displacement field $\mathbf{u}$.

The standard procedure is to eliminate the displacements between the equations repeatedly to produce equa-

tions with only strains as unknowns. First we take two normal components $\epsilon_{(ll)}$ and $\epsilon_{(mm)}$ and the shearing component $\epsilon_{(lm)}$ with the summation convention suspended.

Taking second partial derivatives, we have

$$\epsilon_{(ll,mm)} = u_{(l,lm m)}$$

$$\epsilon_{(mm,ll)} = u_{(m,ml l)}$$

.

$$\begin{aligned}\epsilon_{(lm,lm)} &= \frac{1}{2} (u_{(l,ml m)} + u_{(m,ll m)}) \\ &= \frac{1}{2} (u_{(l,lm m)} + u_{(m,ml l)}).\end{aligned}$$

Next we equate these to get

$$\epsilon_{(ll,mm)} + \epsilon_{(mm,ll)} = 2\epsilon_{(lm,lm)}.$$

Now, taking one normal component $\epsilon_{(ll)}$ and three shearing components ϵ_{lm} , ϵ_{ln} , and ϵ_{nn} and write the second partials

$$\begin{aligned}\epsilon_{(ll,mm)} &= u_{(l,lm n)} \\ \epsilon_{(lm,ln)} &= \frac{1}{2} (u_{(l,ml n)} + u_{(m,ll n)}) \\ \epsilon_{(ln,lm)} &= \frac{1}{2} (u_{(l,nlm)} + u_{(n,ll m)}) \epsilon_{(mn,ll)} = \frac{1}{2} (u_{(m,nll)} + u_{(n,ml l)}).\end{aligned}$$

These can be equated to give

$$\epsilon_{(lm,ln)} + \epsilon_{(ln,lm)} - \epsilon_{(mn,ll)} = \epsilon_{(ll,mm)}.$$

These are six equations relating the components of strain that ensure that they will integrate into a unique displacement field. These can be written explicitly as:

$$\begin{aligned}\epsilon_{11,22} + \epsilon_{22,11} &= 2\epsilon_{12,12} \\ \epsilon_{22,33} + \epsilon_{33,22} &= 2\epsilon_{23,23} \\ \epsilon_{33,11} + \epsilon_{11,33} &= 2\epsilon_{31,31} \\ \epsilon_{12,13} + \epsilon_{13,12} - \epsilon_{23,11} &= \epsilon_{11,23} \\ \epsilon_{23,21} + \epsilon_{21,23} - \epsilon_{31,22} &= \epsilon_{22,31} \\ \epsilon_{31,32} + \epsilon_{32,31} - \epsilon_{12,33} &= \epsilon_{33,12}.\end{aligned}$$

These are known as the *St. Venant* compatibility equations and may also be extracted from the tensor equation

$$\epsilon_{ij,kl} + \epsilon_{kl,ij} = \epsilon_{ik,jl} + \epsilon_{jl,ik}$$

which represents $3^4 = 81$ equations, where only six are distinct.

Lecture 8: Linear isotropic materials and the uniaxial stress

16. Marts 2026

12 Material Behavior

12.1 Uniaxial Behavior

The most elementary and well-known description of material behavior is *Hooke's law*:

$$\sigma_{11} = E\epsilon_{11}$$

where E is the modulus of elasticity or Young's modulus. This also gives rise to stress strain curves as the one shown on [Figure 12.1](#).

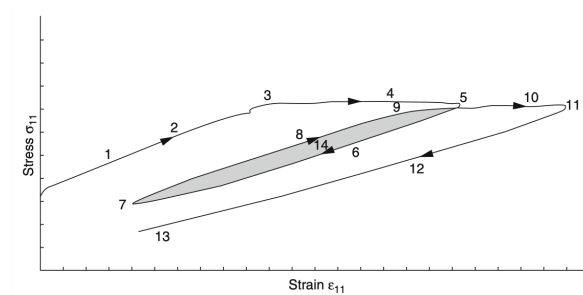


Figure 12.1: Stress-strain curve for two cycles of loading and unloading for mild steel.

Branch (1) is the initial elastic response, from which E may be calculated as the slope. In the vicinity of (2), there is a decrease in the slope, commencing at the *proportional limit* and progressing until the *yield point* is reached at (3). Yielding progresses along (4) until unloading commences at (5) and continues along (6). The unloading ceases and reloading begins at (7) to initiate another cycle that extends through (13) when the test is terminated. Region (14) is known as a *hysteresis loop* and is a measure of the energy dissipated through one excursion beyond the yield point.

12.2 Generalized Hooke's Law

We will now introduce the *strain energy density* W that is a homogeneous quadratic function of the strain components $W(\epsilon_{ij})$. For a body that is slightly strained by gradual application of the loading while the temperature remains constant, a stress component will be produced derivable as

$$\sigma_{ij} = \frac{\partial W}{\partial \epsilon_{ij}}.$$

Here, it is convenient to write W in terms of the engineering strains $\bar{\epsilon}_{ij}$. Thus

$$\sigma_{ij} = \frac{\partial W(\bar{\epsilon}_{ij})}{\partial \bar{\epsilon}_{ij}}$$

where $\bar{\epsilon}_{ij} = (2 - \delta_{ij})\epsilon_{ij}$.

The function should have coefficients such that $W \geq 0$ in order to ensure the stability of the body, with $W(0) = 0$ corresponding to a *natural of zero energy state*.

Now, the generalized Hooke's law is written in the form of a fourth-order tensor

$$\sigma_{ij} = E_{ijkl}\epsilon_{kl}$$

in which the 81 coefficients E_{ijkl} are called the elastic constants. Since W is continuous the order of differentiation does not matter and E_{ijkl} is symmetric. Thus the number of independent equations is reduced from nine to six, the elastic constants from 81 to 36, and then further to $((36 - 6) / 2) + 6 = 21$, when only half of the off-diagonal constants are counted. We represent this relationship in matrix form in terms of the engineering strains as

$$\sigma = E\bar{\epsilon}$$

where

$$\begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{31} \end{Bmatrix} = \begin{bmatrix} E_{1111} & E_{1122} & E_{1133} & E_{1112} & E_{1123} & E_{1131} \\ & E_{2222} & E_{2233} & E_{2212} & E_{2223} & E_{2231} \\ & & E_{3333} & E_{3312} & E_{3323} & E_{3331} \\ & & & E_{1212} & E_{1223} & E_{1231} \\ & & & & E_{2323} & E_{2331} \\ & & & & & E_{3131} \end{bmatrix} \begin{Bmatrix} \bar{\epsilon}_{11} \\ \bar{\epsilon}_{22} \\ \bar{\epsilon}_{33} \\ \bar{\epsilon}_{12} \\ \bar{\epsilon}_{23} \\ \bar{\epsilon}_{31} \end{Bmatrix}.$$

Which can also be written as

$$\begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{31} \end{Bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} \\ & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} \\ & & C_{33} & C_{34} & C_{35} & C_{36} \\ & & & C_{44} & C_{45} & C_{46} \\ & & & & C_{55} & C_{56} \\ & & & & & C_{66} \end{bmatrix} \begin{Bmatrix} \bar{\epsilon}_{11} \\ \bar{\epsilon}_{22} \\ \bar{\epsilon}_{33} \\ \bar{\epsilon}_{12} \\ \bar{\epsilon}_{23} \\ \bar{\epsilon}_{31} \end{Bmatrix}$$

or

$$\sigma = C\bar{\epsilon}.$$

The corresponding strain energy function is

$$\begin{aligned} 2W = & C_{11}\bar{\epsilon}_{11}^2 + 2C_{12}\bar{\epsilon}_{11}\bar{\epsilon}_{22} + 2C_{13}\bar{\epsilon}_{11}\bar{\epsilon}_{33} + 2C_{14}\bar{\epsilon}_{11}\bar{\epsilon}_{12} + 2C_{15}\bar{\epsilon}_{11}\bar{\epsilon}_{23} \\ & + 2C_{16}\bar{\epsilon}_{11}\bar{\epsilon}_{31} + C_{22}\bar{\epsilon}_{22}^2 + 2C_{23}\bar{\epsilon}_{22}\bar{\epsilon}_{33} + 2C_{24}\bar{\epsilon}_{22}\bar{\epsilon}_{12} + 2C_{25}\bar{\epsilon}_{22}\bar{\epsilon}_{23} \\ & + 2C_{26}\bar{\epsilon}_{22}\bar{\epsilon}_{31} + C_{33}\bar{\epsilon}_{33}^2 + 2C_{34}\bar{\epsilon}_{33}\bar{\epsilon}_{12} + 2C_{35}\bar{\epsilon}_{33}\bar{\epsilon}_{23} + 2C_{36}\bar{\epsilon}_{33}\bar{\epsilon}_{31} \\ & + C_{44}\bar{\epsilon}_{12}^2 + 2C_{45}\bar{\epsilon}_{12}\bar{\epsilon}_{23} + 2C_{46}\bar{\epsilon}_{12}\bar{\epsilon}_{31} \\ & + C_{55}\bar{\epsilon}_{23}^2 + 2C_{56}\bar{\epsilon}_{23}\bar{\epsilon}_{31} \\ & + C_{66}\bar{\epsilon}_{31}^2. \end{aligned}$$

The preceding characterization is the most general with which we need to be concerned; such a material is termed *anisotropic*. Fortunately, most engineering materials possess properties of symmetry about one or more of the planes or axes which allow the number of independent constants to be reduced.

This can be reduced for each plane of symmetry for the material ultimately leading to an isotropic material for three-axial symmetry.

The stress-strain relationship can be written in terms of the tensor components ϵ_{ij} using the Lamé constants λ and μ :

$$\begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{31} \end{Bmatrix} = \begin{bmatrix} 2\mu + \lambda & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & 2\mu + \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & 2\mu + \lambda & 0 & 0 & 0 \\ & & & 2\mu & 0 & 0 \\ & & & & 2\mu & 0 \\ & & & & & 2\mu \end{bmatrix} \begin{Bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ \epsilon_{12} \\ \epsilon_{23} \\ \epsilon_{31} \end{Bmatrix}$$

or, in indicial form,

$$\sigma_{ij} = 2\mu\epsilon_{ij} + \lambda\delta_{ij}\epsilon_{kk}.$$

This can also be inverted to express the strains in terms of the stresses:

$$\epsilon_{ij} = \frac{1}{2\mu}\sigma_{ij} - \frac{\lambda}{2\mu(2\mu + 3\lambda)}\delta_{ij}\sigma_{kk}.$$

Although the Lamé constants are perfectly suitable from a mathematical standpoint, it is often more convenient to use engineering material constants that are related to measurements from elementary tests.

For a uniaxial stress with σ_{11} constant and all other $\sigma_{ij} = 0$,

$$\begin{aligned}\sigma_{11} &= (2\mu + \lambda)\epsilon_{11} + \lambda\epsilon_{22} + \lambda\epsilon_{33}, \\ 0 &= \lambda\epsilon_{11} + (2\mu + \lambda)\epsilon_{22} + \lambda\epsilon_{33}, \\ 0 &= \lambda\epsilon_{11} + \lambda\epsilon_{22} + (2\mu + \lambda)\epsilon_{33}.\end{aligned}$$

These can be equated to yield the basic form of Hooke's law:

$$\sigma_{11} = \frac{\mu(2\mu + 3\lambda)}{\mu + \lambda}\epsilon_{11} = E\epsilon_{11}.$$

From the same uniaxial stress state, the *fractional contraction* may be computed as

$$-\frac{\epsilon_{22}}{\epsilon_{11}} = -\frac{\epsilon_{33}}{\epsilon_{11}} = \frac{\lambda}{2(\mu + \lambda)} = \nu$$

where ν is Poisson's ratio.

A third engineering constant is obtained from the state of pure shear in two dimensions, given by $\sigma_{12} = \sigma_{21} =$ constant, all other $\sigma_{ij} = 0$

$$\sigma_{12} = 2\mu\epsilon_{12} = 2G\epsilon_{12}$$

where G is the shear modulus.

Although E , ν , and G are convenient, we realize only two of these are independent since $G = \mu$ and both E and ν are defined in terms of λ and μ . The relationships between the Lamé and engineering constants are collected as:

$$\begin{aligned}\mu &= \frac{E}{2(1 + \nu)} \\ \lambda &= \frac{\nu E}{(1 + \nu)(1 - 2\nu)} \\ E &= \frac{\mu(2\mu + 3\lambda)}{\mu + \lambda} \\ \nu &= \frac{\lambda}{2(\mu + \lambda)} \\ G &= \mu.\end{aligned}$$

Another relationship between the constants is defined as the bulk modulus

$$K = \frac{E}{3(1 + 2\nu)}.$$

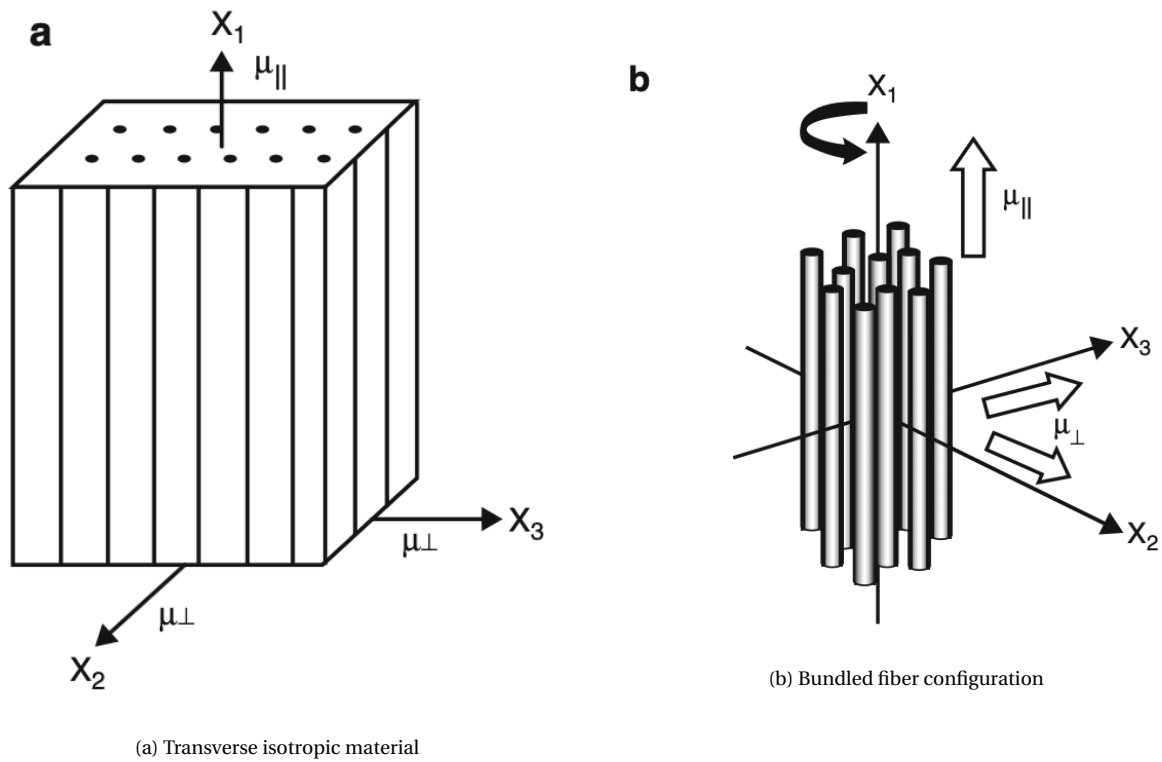


Figure 12.2: Transverse isotropic material and bundled fiber configurations

Note that for λ to remain finite, Poisson's ratio must lie between

$$-1 < \nu < 0.5.$$

Also, for $\nu = 0.5$, the upper limit, the dilation $\Delta = 0$, and the material is said to be incompressible.

The explicit form of the generalized Hooke's law in terms of the engineering constants and tensoral strains is best written as

$$\epsilon = C^{-1} \sigma$$

or

$$\begin{Bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ \epsilon_{12} \\ \epsilon_{23} \\ \epsilon_{31} \end{Bmatrix} = \begin{bmatrix} \frac{1}{E} & -\frac{\nu}{E} & -\frac{\nu}{E} & 0 & 0 & 0 \\ -\frac{\nu}{E} & \frac{1}{E} & -\frac{\nu}{E} & 0 & 0 & 0 \\ -\frac{\nu}{E} & -\frac{\nu}{E} & \frac{1}{E} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2G} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2G} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2G} \end{bmatrix} \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{31} \end{Bmatrix}.$$

The elements of C^{-1} are called *compliances*.

12.3 Transverse Isotropic Material

In Figure 12.2a, a configuration with the fibers oriented along the z -axis, perpendicular to the symmetric $x - y$ -plane is shown. These fibers are able to stretch in a single direction as shown on Figure 12.2b. This might correspond to the muscle fibers inside a muscle.

This is a transverse isotropic material with five elastic constants, $\lambda_{\perp}$, $\lambda_{\parallel}$, λ_M , $\mu_{\perp}$, and $\mu_{\parallel}$. The λ terms are related to the propagation of longitudinal waves in the different directions, while the μ terms are shear moduli. The

expanded stress-strain matrix C_{TI} is:

$$C_{TI} = \begin{bmatrix} 2\mu_{\parallel} + \lambda_{\parallel} & \lambda_M & \lambda_M & 0 & 0 & 0 \\ \lambda_M & 2\mu_{\perp} + \lambda_{\perp} & \lambda_{\perp} & 0 & 0 & 0 \\ \lambda_M & \lambda_{\perp} & 2\mu_{\perp} + \lambda_{\perp} & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu_{\perp} & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu_{\parallel} & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu_{\parallel} \end{bmatrix}.$$

Note, that this is expressed in terms of Lamé constants μ and λ . An alternate form using engineering constants is

$$\begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ 2\epsilon_{23} \\ 2\epsilon_{13} \\ \epsilon_{12} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_t} & -\frac{\nu_{pt}}{E_p} & -\frac{\nu_{pt}}{E_p} & 0 & 0 & 0 \\ -\frac{\nu_{tp}}{E_t} & \frac{1}{E_p} & -\frac{\nu_p}{E_p} & 0 & 0 & 0 \\ -\frac{\nu_{tp}}{E_t} & -\frac{\nu_p}{E_p} & \frac{1}{E_p} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{\mu_p} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{\mu_t} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{\mu_t} \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{bmatrix}.$$

The seven variables in the above compliance matrix are related by the additional equations

$$\frac{\nu_{pt}}{E_p} = \frac{\nu_{tp}}{E_t} \iff \frac{E_t}{E_p} = \frac{\nu_{tp}}{\nu_{pt}}$$

$$\mu_p = \frac{E_p}{2(1 + \nu_p)}.$$

Also, as the compliance matrix should be positive definite, the following five relationships exist for stability

$$\begin{aligned} E_t, E_p, u_t, u_p &> 0 \\ |\nu_p| &< 1 \\ \nu_{pt} &< \sqrt{\frac{E_p}{E_t}} \\ \nu_{tp} &< \sqrt{\frac{E_t}{E_p}} \\ 1 - \nu_p^2 - 2\nu_{pt}\nu_{tp} - 2\nu_{pt}\nu_{tp}\nu_p &> 0. \end{aligned}$$

Lecture 9: Thermal strains

23. Marts 2026

13.1 Thermal Strains

Temperature changes may be a source of stresses in elastic systems. It is reasonable to assume that a linear relationship exists between the temperature difference from the datum value and the corresponding strain

$$\epsilon_{(ii)}(T) = \alpha (T - T_0)$$

in which $\epsilon_{(ii)}(T)$ is the linear strain due to the temperature difference $T - T_0$, T_0 is the datum temperature and α is the coefficient of thermal expansion.

Using this, we can write

$$\epsilon_{ij} = \frac{1}{E} \left((1 + \nu) \sigma_{ij} - \nu \delta_{ij} \sigma_{kk} \right) + \alpha \delta_{ij} (T - T_0)$$

and

$$\sigma_{ij} = 2\mu\epsilon_{ij} + \lambda\delta_{ij}\epsilon_{kk} - \alpha\delta_{ij}(3\lambda + 2\mu)(T - T_0).$$